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Dynamic postural stability indices in athletes: a case-control study on chronic ankle instability during multi-directional landing assessments

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Abstract

Background Our understanding of the impact of chronic ankle instability (CAI) on dynamic postural stability remains limited, primarily due to the absence of reliable and accurate assessment tools. This study aimed to explore two aspects of dynamic postural stability: (1) differences in the dynamic postural stability index (DPSI) and its directional components between athletes with CAI and those with stable ankles; and (2) the impact of jump direction (forward vs. lateral) on these indices across both groups.

Methods We recruited fifteen athletes with CAI and fifteen healthy athletes as controls. The injured side of CAI subjects and the matched side of controls performed forward and lateral single-leg landing tasks (from a 20 cm height onto a force platform, three repetitions). DPSI and its directional indices (anterior-posterior, medial-lateral, and vertical) were calculated according to Wikstrom's method.

Results Athletes with CAI demonstrated significantly greater anterior-posterior stability index (APSI) scores ($F_{(1,28)}=9.864$, $p=0.004$, $\eta^2_p=0.261$) and vertical stability index (VSI) scores ($F_{(1,28)}=6.193$, $p=0.019$, $\eta^2_p=0.181$) during the forward landing task, and significantly greater medial-lateral stability index (MLSI) scores ($F_{(1,28)}=10.144$, $p=0.004$, $\eta^2_p=0.266$) during the lateral landing task. APSI (CAI: $F_{(1,28)}=42.616$, $p<0.001$, $\eta^2_p=0.603$; CON: $F_{(1,28)}=6.229$, $p=0.019$, $\eta^2_p=0.182$) and DPSI (CAI: $F_{(1,28)}=13.062$, $p=0.001$, $\eta^2_p=0.318$; CON: $F_{(1,28)}=9.673$, $p=0.004$, $\eta^2_p=0.257$) scores revealed a significant difference between directions in both groups. MLSI scores exhibited a significant increase in the lateral direction in the CAI group ($F_{(1,28)}=18.289$, $p<0.001$, $\eta^2_p=0.395$).

Conclusions Athletes with CAI demonstrated dynamic stability deficits, and the direction of the jump can affect dynamic postural stability in the sagittal and frontal planes.

Trial registration Chinese Clinical Trial Registry, ChiCTR2200062443, Registered 7 August 2022.

Keywords Ankle injuries, Postural balance, Dynamic postural stability index

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Background

Chronic ankle instability (CAI) is a prevalent condition characterized by recurrent episodes or feelings of "giving way" and instability, along with persistent symptoms such as pain, weakness, or limited range of motion in the ankle joint, as well as self-reported function impairment [1, 2]. Notably, 33%–40% of individuals who experience lateral ankle sprains (LAS) will develop CAI [2, 3]. The incidence of CAI among collegiate athletes (23.4%) is considerably higher compared to that in the general population (0.7%–1.1%) [4]. Moreover, athletes with CAI demonstrate a markedly increased prevalence of osteoarthritis [5, 6], which not only causes significant distress to patients but also imposes a substantial burden on both healthcare and economic systems.

Dynamic postural control deficits represent a defining characteristic of CAI, stemming from impairments in somatosensory and/or motor function [1, 7, 8], and these deficits are particularly pronounced during functional activities like jump landing [9]. The dynamic postural stability index (DPSI) quantifies an individual's capacity to efficiently dissipate ground reaction forces (GRFs) generated during a jump landing [10], integrating lower limb kinematics, muscle activation patterns, and eccentric control [11]. Its directional components—anterior-posterior (APSI), medial-lateral (MLSI), and vertical (VSI)—are calculated along the AP, ML, and vertical axes, respectively [12]. As a composite measure of the GRFs in three planes, DPSI is highly responsive to changes in each direction [13]. Given this, evaluating DPSI and its directional indices during jump landing tasks can serve as a reliable and precise laboratory measure to quantify dynamic postural control in patients with CAI [12], making it pivotal to advancing injury prevention and rehabilitation strategies.

The DPSI has undergone rigorous methodological validation, with feasibility, reliability, and precision studies consistently confirming its high reliability (intraclass correlation coefficient = 0.96) and exceptional precision (SEM = 0.03) [14], establishing it as a sound tool for assessing landing-related postural control. However, existing research predominantly focuses on forward landing, which is inadequate for exploring the causes, prevention, and rehabilitation of lower extremity injuries [10, 13]. This narrow emphasis fails to capture the multi-directional landing requirements inherent in most sports, such as lateral cutting maneuvers in basketball. Consequently, there is currently insufficient data to determine whether there are disparities in DPSI during multi-directional landing among athletes with and without CAI. Most importantly, no studies to date have systematically compared DPSI and its directional components across both forward and lateral landing tasks between athletes with CAI and their healthy counterparts. This leaves unresolved whether CAI-related stability deficits vary by landing direction, and whether directional differences in DPSI are consistent across groups. The present study thus included a comparison between healthy athletes and athletes with CAI, while also incorporating jump-landing tests in various directions.

To address the identified gaps, this study employs a 2 (groups: CAI vs. CON) × 2 (directions: forward vs. lateral) factorial design to elucidate the impact of ankle joint stability and landing direction on dynamic postural control. Guided by prior research [10, 12, 15], we hypothesized that (1) compared to the CON group, athletes with CAI would exhibit higher DPSI, APSI, and VSI scores during forward jump landing tasks, and higher DPSI, MLSI, and VSI scores during lateral jump landing tasks; and (2) forward and lateral jump landing tasks would differentially influence the three-dimensional components of the DPSI—specifically, forward jump landing tasks would yield higher APSI scores, while lateral jump landing tasks would yield higher MLSI scores.

Methods

Study design

This case-control study is a component of a comprehensive research project aimed at investigating the potential mechanisms underlying CAI. The current study incorporates one between-group factor (group) and one within-group factor (direction). All tests were performed only on the injured side in CAI individuals or the matched side in healthy controls, and comparisons were made between groups and directions. Based on the findings of previous similar studies [12, 16], we used G*Power software (version 3.1) to determine the required sample size, with a power of 0.90, a Type I error rate of 0.05, and an effect size f = 0.40. This calculation indicated that at least 10 participants per group would be necessary.

Participants

A total of 30 volunteers from Beijing Sport University were enrolled in this study. All collegiate athletes were required to meet China's national second-level athlete standard—defined as an athlete with solid skills who has achieved specific results in national or provincial sports competitions or reached designated performance standards—and to actively practice and/or compete at least three times per week for two hours. Only individuals with unilateral ankle sprains were included. More inclusion criteria for CAI were based on the recommendations provided by the International Ankle Consortium [17]. Additional criteria for healthy controls included no history of ankle sprain and a CAIT score ≥ 28. Individuals were excluded if they had any of the following: a history of prior surgeries on the lower limb musculoskeletal system, chronic conditions affecting the musculoskeletal

structures that could lead to postural control issues, or recent acute injuries (within the last three months) to the lower limb musculoskeletal components [17, 18].

All participants provided written informed consent. The study received approval from the Sports Science Experiment Ethics Committee of Beijing Sport University (approval number: 2022058 H), adhered to the Declaration of Helsinki, and was registered in the Chinese Clinical Trial Registry (registration number: ChiCTR2200062443).

Test protocol

All the tests were carried out at the Sports Medicine Rehabilitation Center of Beijing Sport University. Participants completed the test while wearing comfortable sportswear without shoes. Each participant engaged in a 10-minute preparatory routine that included running and stretching activities designed to target the muscles around the ankle joint. Subsequently, subjects were instructed to perform multi-directional jump landing tasks involving both forward and lateral directions. Only the injured side of CAI subjects and the matched side of healthy subjects were tested. A specially customized voice package was used to remind the subjects to start in each trial. Upon receiving the audio cue, participants initiated the movement by pushing off with both feet and landing on the designated testing leg from a 20 cm high aerobic step, which was placed 5 cm behind a force plate [19]. They were instructed to maintain balance for 8 s while keeping their hands on their hips, the opposite hip flexed at 30°, and the knee bent at 45°. Additionally, they were required to keep their eyes fixed on a visual target placed on the wall. No specific instructions for jump height were given. Three formal assessments were administered, and each participant was given at least one practice opportunity before the formal evaluations. If a participant was unable to maintain balance for the entire required duration, removed their hands from their hips, hopped, shifted the involved foot on the force plate, touched the ground with the opposite leg, or landed with partial contact of the involved foot on the force plate, the trial was deemed unsuccessful. In the event of failure, participants were to stop and retry the test until they achieved three successful trials in each direction. This procedure was conducted for both landing directions, with a 1-minute rest period between them.

Data processing

For DPSI, we used a force-platform (Kistler.9286 A, Kistler Instrument AG, Winterhur, Switzerland) at a sampling rate of 1000 Hz to collect GRF data. To define the global reference system, the anterior-posterior direction was aligned with the y-axis, the medial-lateral direction was assigned to the x-axis, and the vertical direction was positioned along the z-axis. All force plate measurements were converted from analog to digital format and subsequently stored on a personal computer utilizing Bioware software. (version 5.4.3, Kistler Instrument AG, Winterthur, Switzerland). The raw data were subsequently exported into comma delimited (CSV) files and processed using Python (version 3.9, Python Software Foundation, Wilmington, USA). The data was first imported into Python and initially processed by applying a 12 Hz second-order low-pass Butterworth filter for rectification and filtering [20]. Then the first 5 s of filtered data after initial contact were analyzed, and initial contact was defined as the time when the vertical GRF was greater than 10 N. DPSI and its directional indices were calculated using a previously published formula (see Table 1), which quantifies the standard deviation of GRF fluctuations around their respective reference points. The GRFs in the formula are first normalized by body weight to allow inter-subject comparisons, and the index is subsequently obtained by dividing the sum of squared deviations by the total number of data points in the trial [13]. Thus, higher scores suggest increased variability [13].

Data analysis

Data analysis was performed using SPSS software version 24.0 (IBM Corp., Armonk, NY). The Shapiro-Wilk (S-W) test and Levene test were employed to assess the normality of the variables and the homogeneity of variances, respectively. Since the data satisfied these assumptions, parametric statistical methods were utilized in subsequent analyses. Independent-samples t-tests were used to evaluate the differences between groups in demographic characteristics. A repeated-measures ANOVA was conducted to examine DPSI and its three-dimensional

Table 1 DPSI formulas

$$\mathrm{MLSI} = \sqrt{\left[\sum \left(\{0 - \mathrm{F_y}\}/\mathrm{BW}\right)^2\right]}/\mathrm{S}$$

$$\mathrm{APSI} = \sqrt{\left[\sum \left(\{0 - \mathrm{F_x}\}/\mathrm{BW}\right)^2\right]}/\mathrm{S}$$

$$\mathrm{VSI} = \sqrt{\left[\sum \left(\{\mathrm{BW} - \mathrm{F_z}\}/\mathrm{BW}\right)^2\right]}/\mathrm{S}$$

$$\mathrm{DPSI} = \sqrt{\left[\sum \left(\{0 - \mathrm{F_y}\}/\mathrm{BW}\right)^2 + \sum \left(\{0 - \mathrm{F_x}\}/\mathrm{BW}\right)^2 + \sum \left(\{\mathrm{BW} - \mathrm{F_z}\}/\mathrm{BW}\right)^2\right]}/\mathrm{S}$$

Fx, Fy, Fz, force in the x, y, and z plane, respectively

MLSI Medial-lateral stability index, *APSI* Anterior-posterior stability index, *VSI* Vertical stability index, *DPSI* Dynamic postural stability index, *BW* Body weight, *S* Number of samples

indices, incorporating one between-subjects factor (group) and one within-subjects factor (direction). Means and standard deviations (SDs) were calculated for all continuous variables. Partial eta squared (η^2_p) values were provided to reflect the effect sizes. Effect sizes below 0.01 were considered small, those ranging from 0.01 to 0.06 were deemed medium, and values of 0.14 or higher indicated a large effect [21]. The statistical significance level was set at $p < 0.05$.

Result

No significant differences were noted between the groups regarding demographic characteristics, except for the CAIT scores (see Table 2).

In the forward landing direction, individuals in the CAI group exhibited significantly higher APSI ($F_{(1,28)} = 9.864$, $p = 0.004$, $\eta^2_p = 0.261$) and VSI scores ($F_{(1,28)} = 6.193$, $p = 0.019$, $\eta^2_p = 0.181$) compared to controls. For lateral landings, the CAI group showed markedly elevated MLSI scores relative to controls ($F_{(1,28)} = 10.144$, $p = 0.004$, $\eta^2_p = 0.266$) (see Table 3).

Table 2 Participant demographics

Measure	CAI	CON	t	*p*	d
Gender, n(M/F)	10/5	10/5			
Age, y	20.07 ± 0.88	20.87 ± 1.68	−1.629	0.115	−0.597
Height, cm	174.50 ± 8.10	175.19 ± 8.24	−0.232	0.818	−0.084
Mass, kg	66.38 ± 11.59	67.93 ± 9.60	−0.399	0.693	−0.146
CAIT, score	18.93 ± 3.71	29.47 ± 0.52	−10.885	0.000*	−3.979

CAI Chronic ankle instability group, *CON* Healthy control group, *M* Male, *F* Female, *CAIT* Cumberland Ankle Instability Tool

*Indicates p < 0.05

Table 3 DPSI outcomes

	Group, Mean ± SDs		F	*p*	η^2_p
	CAI	CON			
Forward landing task					
APSI	0.414 ± 0.084	0.325 ± 0.069	9.864	0.004*	0.261
MLSI	0.119 ± 0.015	0.135 ± 0.028	3.523	0.071	0.112
VSI	0.091 ± 0.004	0.087 ± 0.005	6.193	0.019*	0.181
DPSI	0.256 ± 0.065	0.247 ± 0.057	0.181	0.674	0.006
Lateral landing task					
APSI	0.259 ± 0.048†	0.266 ± 0.040†	0.179	0.675	0.006
MLSI	0.155 ± 0.024‡	0.128 ± 0.022	10.144	0.004*	0.266
VSI	0.090 ± 0.003	0.089 ± 0.005	0.347	0.561	0.012
DPSI	0.192 ± 0.041§	0.191 ± 0.030§	0.001	0.972	0.000

CAI Chronic ankle instability group, *CON* Healthy control group, *APSI* Anterior-posterior stability index, *MLSI* Medial-lateral stability index, *VSI* Vertical stability index *DPSI* Dynamic postural stability index

* Indicates p < 0.05 between groups

†Indicates a statistically significant difference (p < 0.05) between the forward and lateral landing tasks in APSI

‡Indicates a statistically significant difference (p < 0.05) between the forward and lateral landing tasks in MLSI

§Indicates a statistically significant difference (p < 0.05) between the forward and lateral landing tasks in DPSI

Within the CAI group, landing direction exerted a statistically significant effect on the APSI ($F_{(1,28)} = 42.616$, $p < 0.001$, $\eta^2_p = 0.603$), MLSI ($F_{(1,28)} = 18.289$, $p < 0.001$, $\eta^2_p = 0.395$) and DPSI ($F_{(1,28)} = 13.062$, $p = 0.001$, $\eta^2_p = 0.318$) scores. Specifically, APSI and DPSI scores were significantly higher during forward landings, whereas MLSI scores were significantly higher during lateral landings. In contrast, no significant difference in VSI ($F_{(1,28)} = 0.822$, $p = 0.372$, $\eta^2_p = 0.029$) scores was detected between the two landing directions. For the CON group, a similar pattern emerged, with landing direction significantly influencing both APSI ($F_{(1,28)} = 6.229$, $p = 0.019$, $\eta^2_p = 0.182$) and DPSI ($F_{(1,28)} = 9.673$, $p = 0.004$, $\eta^2_p = 0.257$) scores. Consistent with the CAI group, VSI scores did not differ significantly between forward and lateral landings ($F_{(1,28)} = 1.462$, $p = 0.237$, $\eta^2_p = 0.050$) (see Table 3).

Discussion

This study aims to explore possible differences in dynamic postural control between athletes with and without CAI, taking into account the roles of ankle stability and landing direction. The results provide partial support for our hypotheses. For Hypothesis 1, the CAI group demonstrated significantly higher APSI and VSI scores during forward landings and higher MLSI scores during lateral landings, aligning with predictions. However, no significant between-group difference was observed in DPSI, and the CAI group did not show elevated VSI during lateral landings, thus failing to fully support the hypothesis. For Hypothesis 2, landing direction differentially influenced APSI and MLSI—forward landings were associated with higher APSI, whereas lateral landings yielded higher MLSI—consistent with predictions. Nevertheless, landing direction did not affect VSI, a critical component of three-dimensional stability, resulting in only partial confirmation of the hypothesis.

Previous studies have consistently reported that patients with CAI exhibit reduced postural stability compared to healthy controls. Specifically, the CAI group demonstrated significantly higher scores in both the anterior-posterior and vertical axes, as well as in the overall DPSI, when performing a forward jump-landing protocol [10, 12, 13]. This is partially consistent with our findings: we similarly observed significantly elevated APSI and VSI in the CAI group relative to healthy controls, reflecting impaired anterior-posterior and vertical postural control in these patients. However, unlike prior studies, no significant difference in overall DPSI was detected between the CAI and control groups. Notably, the forward jump landing primarily involves a sagittal plane motion pattern that appears to generate insufficient force to elicit intergroup disparities in the frontal plane. This may explain why no significant difference in MLSI was observed between the CAI and CON groups, as the

task's mechanical demands failed to challenge the frontal plane control mechanisms that are typically impaired in CAI. With respect to APSI, Wikstrom's research reported that both the Copers and CAI groups had higher APSI scores compared to the healthy control group, directly linking ankle injury history to anterior-posterior postural control deficits [13]. These findings are corroborated by time to stabilization (TTS) measurements, another key indicator of dynamic postural control. Previous research has indicated significant differences in TTS in the anterior-posterior direction among different groups. Specifically, individuals in the CAI or FAI groups require more time to regain a stable level of GRFs [11, 22, 23]. Similarly, our study revealed that the CAI group generated significantly higher APSI scores when performing the forward landing task. It is proposed that the differences in APSI observed between the uninjured control group and the CAI group are attributed to impairments in feedback neuromuscular control [24]. Specifically, ankle ligament injury causes deficits in mechanoreceptor signaling, leading to articular deafferentation and delayed reflex responses of evertor muscles, which prolongs center of mass (COM) stabilization in the anterior-posterior direction [20, 25, 26]. Despite potential differences in landing task parameters across studies, individuals may adopt task-specific control strategies; however, the core deficit of impaired neuromuscular feedback persists. We suggest that the symptoms associated with the initial trauma were not eliminated and may have even worsened in the CAI group due to frequent participation in sports and high-intensity athletic activities, which exacerbate neuromuscular control impairments.

Most prior research has used only forward jump-landing protocols, which may fail to capture the critical neuromuscular control and dynamic postural stability mechanisms relevant to lateral ankle pathologies. This limitation is notable because LAS predominantly occur in the frontal plane. To address this gap, the present study employed a lateral jump-landing paradigm: this task not only effectively replicates the biomechanical conditions that cause LAS (directly challenging the ankle's vulnerable structures) [24], but also allows us to assess dynamic postural stability and identify novel insights into CAI-related functional deficits. Biomechanically, the body's COM primarily oscillates in the frontal plane during lateral jump landings. This key characteristic provides a plausible mechanistic explanation for the observed results regarding the MLSI— a metric sensitive to frontal-plane balance control. Specifically, athletes with CAI demonstrated significantly higher MLSI scores than healthy controls during lateral landing trials in the present study. This finding aligns with prior research showing that individuals with CAI frequently use a proximal-dominant "hip strategy" to manage movement [27]. Given that this compensatory strategy is less efficient at balancing control and may result in higher shear forces between the body and the ground, it likely accounts for the elevated MLSI scores in the CAI group. However, Shiravi et al. found no differences between athletes with and without CAI during the lateral landing protocol in DPSI and its directional indices. Nonetheless, MLSI scores showed a significant increase compared to APSI and VSI scores [16]. Our findings present a contrasting result, as the APSI scores exhibited higher values compared to MLSI scores, a result consistent with conclusions from others [12, 15]. Additionally, while those studies identified VSI as the highest among stability indices—another discrepancy from our results—minor variations in jump-landing protocols may explain these inconsistencies. In our experiment, participants jumped from a 20 cm height; this likely produced lower vertical impulse than the take-off protocols used by Wikstrom and Brown, which may have reduced VSI scores relative to other indices.

The absence of a significant difference in the DPSI between our CAI and control groups stands in contrast to prior findings [10, 12, 13]. A critical distinction lies in the jump-landing protocols, which directly impact DPSI sensitivity by shaping stability demands. Both Wikstrom et al. and Brown et al. [10, 12, 13] adopted the protocol first described by Ross and Guskiewicz: participants jumped bipedally to touch an overhead marker (set at 50% of their maximum vertical leap) before landing unipedally on a force plate. Brown et al. further reported that this marker height ranged from 15.3 to 17.9 cm across directions—lower than the 20 cm in our study. Even so, this protocol generates greater momentum and higher landing impact, amplifying postural control challenges and making DPSI more likely to detect CAI-related deficits. This distinction is meaningful given that rapid post-landing stabilization is a well-recognized protective characteristic [28]. An increase in DPSI and its directional components suggests that individuals take a longer time to stabilize their COM from a dynamic to static state due to ankle sprain trauma [13]. Notably, DPSI's limited sensitivity in our study also reflects its composite nature: single-component indices (e.g., APSI, MLSI) are more responsive to direction-specific stability fluctuations. These components not only enable precise identification of postural weak directions in CAI athletes but also provide targeted rehabilitation targets—addressing a gap in the aggregated DPSI's ability to capture subtle, directionally specific deficits.

Additionally, we have observed a significant impact of jump direction on the DPSI and its three-dimensional components, with this specificity aligning with biomechanical principles: forward landings primarily challenge sagittal-plane balance (driving APSI and composite DPSI elevations), while lateral landings target frontal-plane stability (inducing MLSI increases); notably, no significant

differences were noted for VSI. This link between sagittal-plane balance (APSI) and ankle biomechanics is supported by Mineta et al. [29], who found that reduced sagittal-plane ankle energy absorption (a key deficit in CAI patients) correlates with increased APSI. This finding directly aligns with our observation that forward landings (which impose greater sagittal-plane demands on the ankle) elevate APSI. They further noted that MLSI is associated with both ankle energy absorption and clinical function, reinforcing the interplay between lower extremity biomechanics and multiplanar stability. Our results extend prior literature by emphasizing athletic status as a key moderator of directional effects on stability indices. Wikstrom et al. [15] reported that in non-athletes, lateral/diagonal jump-landings raised MLSI scores and forward jump-landings enhanced VSI scores; yet in our athletic sample, we found no significant VSI directional differences. This discrepancy supports the idea that athletes' accumulated movement experience reduces VSI's directional sensitivity, refining the understanding of how landing direction interacts with participant characteristics to influence stability outcomes. Therefore, a single-direction landing task may obscure crucial aspects of neuromuscular control and dynamic postural stability mechanisms. Utilizing multi-directional jump-landings can take into account the effect of landing direction on dynamic postural control, thereby addressing limitations of single-direction tasks in assessing the causation and prevention of lower extremity injuries [30].

Furthermore, there are also some limitations. Firstly, Copers (individuals with a history of ankle sprains who did not progress to CAI) were not selected as the control group in this study. Previous research has emphasized that Copers are more appropriate controls for CAI studies compared to healthy populations without any ankle sprain history, as they share the key exposure factor of ankle sprain experience while differing only in the outcome of CAI development [6]. Excluding Copers may limit our ability to isolate the specific mechanisms underlying ankle instability—for instance, it becomes difficult to distinguish whether observed postural control differences are due to CAI itself or simply the residual effects of a past sprain. Therefore, incorporating Copers alongside CAI patients and healthy controls in future studies is recommended to better elucidate the pathological mechanisms of instability. Second, jumping from a fixed height of 20 cm may not adequately challenge dynamic stability across all participants. Dynamic stability challenges are closely related to an individual's anthropometric characteristics (e.g., lower limb length, center of mass height), and a fixed box height fails to account for inter-participant differences in these traits. Future investigations could normalize box height as a percentage of each participant's height to ensure uniform challenge intensity and enhance outcome sensitivity.

Conclusion

The current study demonstrates that athletes with CAI experience significant deficits in dynamic stability compared to healthy controls. Additionally, the findings indicate that jump directions exert a statistically significant influence on dynamic postural stability in both the sagittal and frontal planes.

Abbreviations

APSI	Anterior posterior stability index
CAI	Chronic ankle instability
COM	Center of mass
DPSI	Dynamic postural stability index
LAS	Lateral ankle sprains
MLSI	Medial-lateral stability index
GRFs	Ground reaction forces
TTS	Time to stabilization
VSI	Vertical stability index

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Supplementary Information

All methods were performed in accordance with the relevant guidelines and regulations.

Authors' contributions

ZX, RB, and GQ contributed to conceptualization; ZX, and RB contributed to methodology; ZX participated in software; ZX and RB done the formal analysis and investigation; ZX contributed to data curation; ZX done the writing—original draft preparation and writing—review and editing; ZX, RB and GQ provided the resources and project administration; GQ supervised the study. All authors have read and approved the final version of the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Sports Science Experiment of Beijing Sport University (2022058 H). The written informed consent was obtained from all participants.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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